

Assume $X \in \mathbb{R}^{n \times p}$

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disc 4

n observations
 p features.

X has been mean centered.

$$X = U \Sigma V^T$$

U, V
are
orthonormal

PC
scores

$V =$ directions
in original feature
space

Then the $p \times p$ covariance matrix is
given by

$$S = \frac{1}{n-1} X^T X$$

$X^T X$ is symmetric

$$(X^T X)^T = X^T (X^T)^T = X^T X$$

$\Rightarrow S$ is symmetric.

plug the SVD into $X^T X$

$$S = \frac{1}{n-1} (U \Sigma V^T)^T (U \Sigma V^T)$$

$$= \frac{1}{n-1} V \Sigma U^T U \Sigma V^T$$

$$U^T U = I$$

$$V^T V = I$$

$$= \frac{1}{n-1} V \Sigma^2 V^T$$

$$= V \frac{\Sigma^2}{n-1} V^T$$

columns of V
 $=$ eigenvectors
of
covariance

variance
along PC_j

$$= \lambda_j = \sigma_j^2 / (n-1) \quad \text{eigenvalues of covariance matrix} = \text{PC directions}$$



Now we know the PC directions are the columns of V . We project X onto V :

$$Z = XV$$

Substitute the SVD:

$$Z = (U \Sigma V^T) V$$

$$V^T V = I$$

$$Z = U \Sigma$$

Z_{ij} = score of observation i on PC_j

↓ since $ZV^T = XVV^T = X$

$$X_i = Z_{i1} V_1^T + Z_{i2} V_2^T + \dots$$

V : new PCA direction

XV : express observations in PCA coordinates.

$(XV)V^T$: rotates them back to original feature coordinates

$$X = U \Sigma V^T \quad \text{data} = \underbrace{U \Sigma}_{\text{PC scores}} \times \underbrace{V^T}_{\text{PC directions}}$$